Hyperbolic Functions: Domain and Range

Questions 1–10

Question 1. For each of the following functions, sketch the graph and state whether it is one-to-one or if it is many-to-one.

a) $y = \sinh x$
b) $y = \cosh x$
c) $y = \tanh x$

The answers for the following questions which require a range of values should be given in set notation.

Question 2. State the largest possible domain and range of the following functions:

a) $y = \sinh x$
b) $y = \cosh x$
c) $y = \tanh x$

Question 3. State the range of the following, given $x < 0$:

a) $y = \sinh x$
b) $y = \cosh x$
c) $y = \tanh x$

Question 4. State the range of the following, given $x \in \mathbb{R}$:

a) $f(x) = \sinh^2 x$
b) $y = \cosh x + 3$
c) $g(x) = \tanh(x - 2)$
d) $y = 2\sinh x$
e) $h(x) = -\cosh^2 x$
f) $y = \tanh^2 x + 5$

Question 5.

a) Write $2\sinh x + 4\cosh x$ in the form $ae^x + be^{-x}$ where a and b are real constants.

b) Hence explain why, for $x \in \mathbb{R}$, that $2\sinh x + 4\cosh x > e^{-x}$.

Question 6. Show clearly that, for $x \in \mathbb{R}$,

$$4\sinh^2 x - e^{-2x} > -2.$$

Prerequisite knowledge for Q7: Hyperbolic Double-Angle Formulae

Question 7.

a) Write $\sinh 2x$ in terms of $\sinh x$ and $\cosh x$.

b) Given $x > 0$, write down the range of:

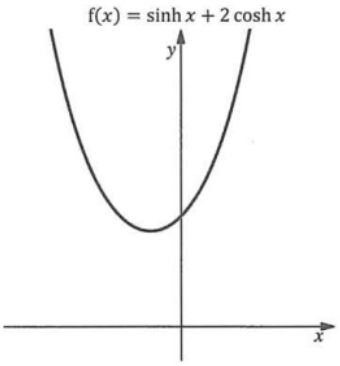
i) $y = \sinh x$

ii) $y = \cosh x$

c) Hence show that $\sinh 2x > \sinh x$ for $x > 0$.

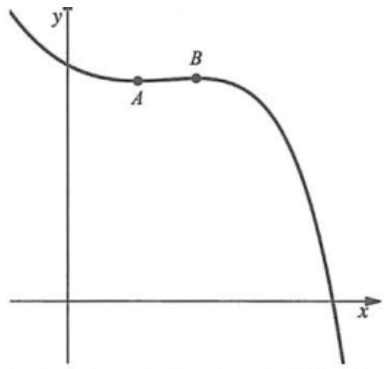
Prerequisite knowledge for Q8–Q10: Differentiation of Hyperbolic Functions

Question 8. The graph below shows $f(x) = \sinh x + 2\cosh x$.



Find the exact largest possible range of f, writing your answer in set notation.

Question 9. The graph below shows the curve with equation $y = 3\cosh x - \sinh x \cosh x$. The points A and B on the curve are stationary points and have x-coordinates a and b respectively.



It is given that the function f is defined as $f(x) = 3\cosh x - \sinh x \cosh x$ where $x \in [a, b]$. Find the range of f in interval notation.

Question 10. It is given that

$$f(x) = \frac{e^{2x} + e^{-2x}}{(e^x + e^{-x})^2} - \tanh^2 x, \qquad x \in \mathbb{R}.$$

a) Show that $f(x) = 2(e^x + e^{-x})^{-2}$, $x \in \mathbb{R}$.
b) Find the coordinates of the stationary point of f.
c) State the range of f, giving your answer in set notation.